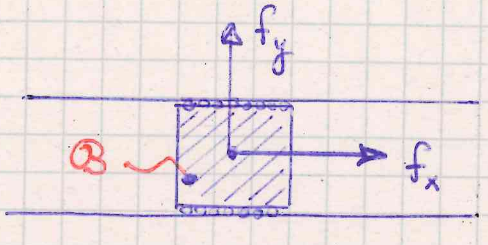
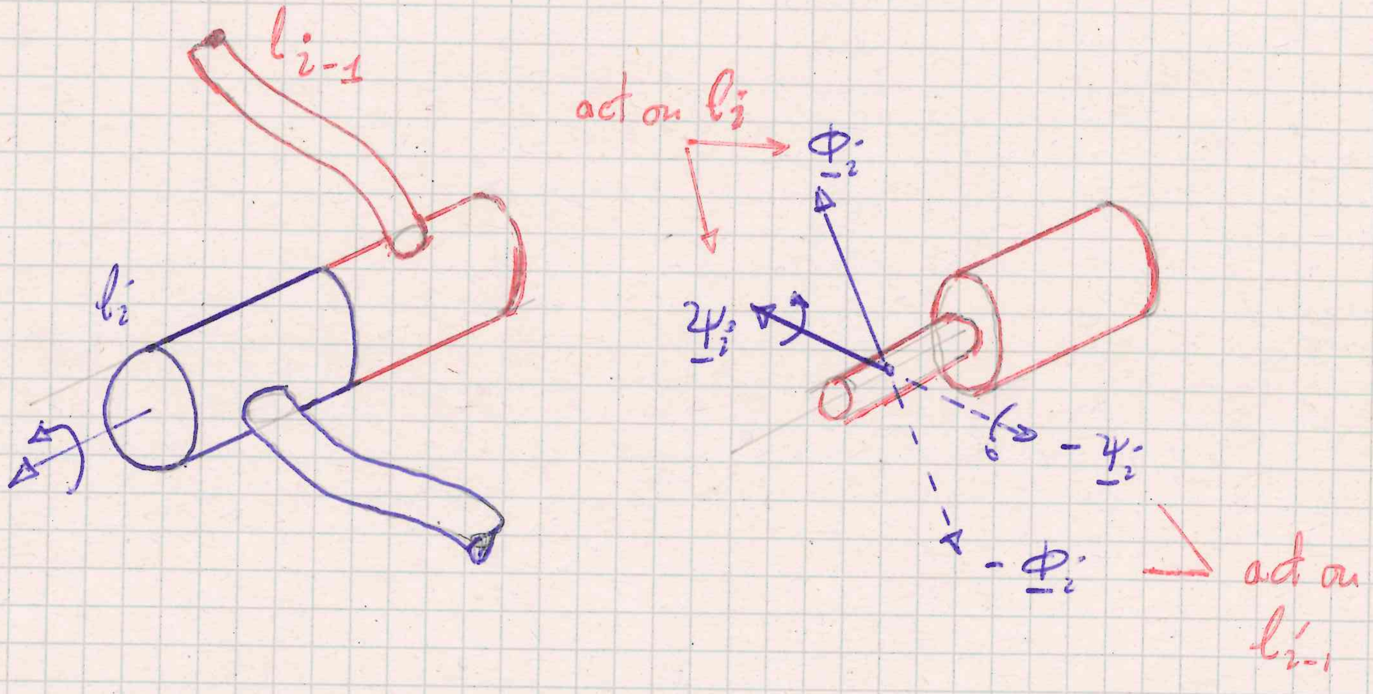
REACTION FORCES / TORQUES



Constraints on body $\mathcal{B}$

$y = y_0$ (constant)
$\vartheta = \vartheta_0$ (constant)

$$\ddot{y} = 0 \iff f_y + \phi_y = 0 \Rightarrow \phi_y = -f_y$$
$$\ddot{\vartheta} = 0 \iff m_z + \Psi_z = 0 \Rightarrow \Psi_z = -m_z$$



VIRTUAL WORK OF REACTION FORCES/TORQUES

Reaction forces/torques express themselfes in joint way on both bodies forming a kinematic pair. It is the convenient (also in view of the forthcoming discussions) to compute the total virtual work of reaction and counter-reaction forces/torques.

Let $\underline{\Phi}_i, \underline{\Psi}_i$ the reaction f/t generated by joint i acting on link l_i and expressed at point P_i

$-\underline{\Phi}_i, -\underline{\Psi}_i$ the counter-reaction f/t generated by joint i acting on link l_{i-1} and expressed at point Q_{i-1}

Then the total virtual work is

$$\delta\Lambda_i = \overbrace{\underline{\Phi}_i \cdot \delta\underline{P}_i + \underline{\Psi}_i \cdot \delta\underline{\theta}_i}^{\text{v. work on } l_i} + \overbrace{\left(-\underline{\Phi}_i \cdot \delta\underline{Q}_{i-1} - \underline{\Psi}_i \cdot \delta\underline{\theta}_{i-1}\right)}^{\text{v. work on } l_{i-1}}$$

virtual work of reaction f/t due to joint i

$$= \underline{\Phi}_i \cdot \underbrace{\left[\delta\underline{P}_i - \delta\underline{Q}_{i-1}\right]}_{\begin{cases} \delta\underline{z}_{i/i-1} = \underline{K}_i\,\delta q_i & i\in TJ \\ \underline{0} & i \in RJ\end{cases}} + \underline{\Psi}_i \cdot \underbrace{\left[\delta\underline{\theta}_i - \delta\underline{\theta}_{i-1}\right]}_{\begin{cases} \underline{0} & i\in TJ \\ \underline{K}_i\,\delta q_i & i \in RJ\end{cases}}$$

IDEAL JOINTS A joint is ideal if $\delta\Lambda_i = 0$

$$\delta\Lambda_i = \begin{cases} \underline{\Phi}_i \cdot \underline{K}_i\,\delta q_i & i\in TJ \\ \underline{\Psi}_i \cdot \underline{K}_i\,\delta q_i & i\in RJ \end{cases}$$

If joint i is ideal $\delta\Lambda_i = 0$ then:

$i \in TJ \iff \delta\Lambda_i = \underline{\Phi}_i \cdot \underline{K}_i\,\delta q_i = 0 \;\; \forall \delta q_i \iff \underline{\Phi}_i \perp \underline{K}_i$

$i \in RJ \implies \delta\Lambda_i = \underline{\Psi}_i \cdot \underline{K}_i\,\delta q_i = 0 \;\; \forall \delta q_i \iff \underline{\Psi}_i \perp \underline{K}_i$

NOTE The ideal joint model (for kinematic pairs) 17/
is equivalent to the assumption of
frictionless joint (*) -

NOTE In practice friction always exists friction
forces/torques are embedded in actuation terms

NOTE Virtual work of reaction F/T on different points

$$\delta \underline{\Lambda}_i = \underbrace{\underline{\Phi}_i \, \delta \underline{P}_i + \underline{\Psi}_i \, \delta \underline{\theta}_i}_{\delta \underline{\Lambda}_i^{(i)}} + \underbrace{\left[- \underline{\Phi}_i \cdot \delta \underline{Q}_{i-1} - \underline{\Psi}_i \cdot \delta \underline{\theta}_{i-1} \right]}_{\delta \underline{\Lambda}_i^{(i-1)}}$$

$\delta \underline{\Lambda}_i^{(i)}$ virtual work of $\underline{\Phi}_i, \underline{\Psi}_i$ on link l_i as computed at P_i

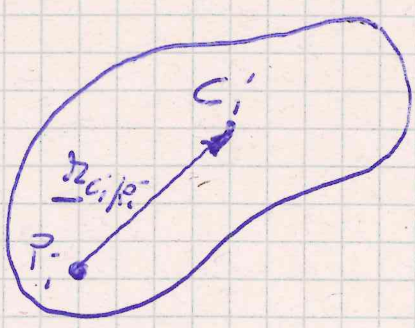
$\delta \underline{\Lambda}_i^{(i-1)}$ virtual work of $-\underline{\Phi}_i, -\underline{\Psi}_i$ on link l_{i-1} as computed at Q_{i-1}

Let us compute $\delta \underline{\Lambda}_i^{(C_i)}$ (the virtual work of $\underline{\Phi}_i, \underline{\Psi}_i$ on link l_i as computed at C_i)

$$\delta \underline{\Lambda}_i^{(C_i)} = \underline{\Phi}_i \cdot \delta \underline{C}_i + \left[\underline{\Psi}_i + \underline{r}_{P_i/C_i} \times \underline{\Phi}_i \right] \cdot \delta \underline{\theta}_i = (*)$$

$$\delta \underline{C}_i = \delta \underline{P}_i + \delta \underline{\theta}_i \times \underline{r}_{C_i/P_i}$$

$$\underline{r}_{C_i/P_i} = - \underline{r}_{P_i/C_i}$$



(*) The non slipping wheel w/ friction is not holonomic but ideal because the friction is not making any mechanical work

$$(*) = \underline{\Phi}_i \cdot \left[\delta \underline{P}_i + \delta \underline{\theta}_i \times \underline{r}_{C_i/P_i}\right] + \left[\underline{\Psi}_i + \underline{r}_{P_i/C_i} \times \underline{\Phi}_i\right] \cdot \delta \underline{\theta}_i \qquad 17.1$$

$$= \underline{\Phi}_i \cdot \delta \underline{P}_i + \underbrace{\underline{\Phi}_i \cdot \left[\delta \underline{\theta}_i \times \underline{r}_{C_i/P_i}\right]}_{\substack{\text{triple product:} \\ \text{make a permutation...}}} + \left[\underline{\Psi}_i + \underline{r}_{P_i/C_i} \times \underline{\Phi}_i\right] \delta \underline{\theta}_i =$$

$$= \underline{\Phi}_i \cdot \delta \underline{P}_i + \left[\underbrace{\underline{r}_{C_i/P_i}}_{-\underline{r}_{P_i/C_i}} \times \underline{\Phi}_i\right] \cdot \delta \underline{\theta}_i + \underline{\Psi}_i \cdot \delta \underline{\theta}_i + \left[\underline{r}_{P_i/C_i} \times \underline{\Phi}_i\right] \cdot \delta \underline{\theta}_i$$

$$= \underline{\Phi}_i \cdot \delta \underline{P}_i + \underline{\Psi}_i \cdot \delta \underline{\theta}_i = \delta \underline{\Lambda}_i^{(i)}$$

Using the same argument we have that

$$\delta \underline{\Lambda}_i^{(i-1)} = \delta \underline{\Lambda}_i^{(C_i-1)}$$

then, said $\underline{\Phi}_i^{TOT}$, $\underline{\Psi}_i^{TOT}$ the total contribution of the

we always have

$$\underbrace{\sum_1^n{}_i \, \delta \underline{\Lambda}_i}_{\sum_1^n{}_i \, \delta \underline{\Lambda}_i^{(C_i)\,TOT}} = \sum_1^n{}_i \, \underline{\Phi}_i^{TOT} \cdot \delta \underline{C}_i + \underline{\Psi}_i^{TOT} \cdot \delta \underline{\theta}_i = 0$$

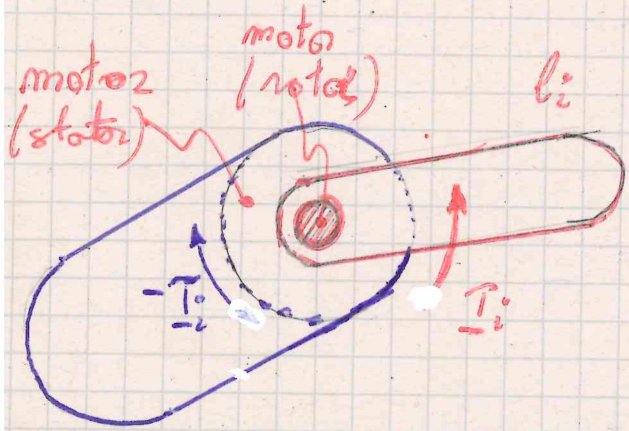
Virtual work principle

ACTUATION FORCES/TORQUES

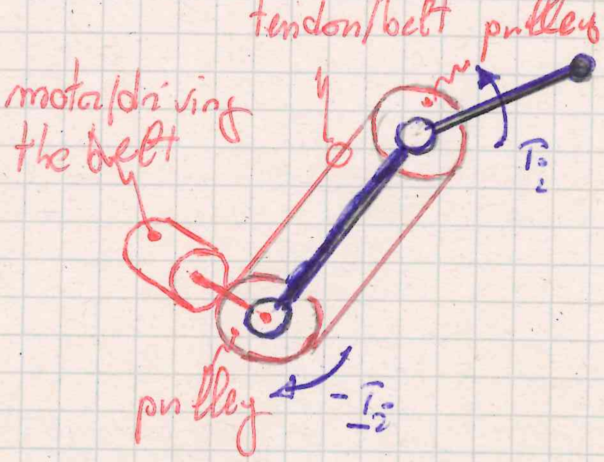
Actuation f/t are active entities generated by actuators (e.g. motors).

Actuation f/t can be

- congruent with a joint (*)
- non congruent with a joint



Congruent actuation



Non congruent actuation

$\boxed{\text{RJ}}$ $\underline{m}_i$: torque generated by actuator i on link l_i

$-\underline{m}_i$: torque generated by actuator i on link l_{i-1}

$$\delta L_i = \underline{m}_i \cdot \delta\underline{\theta}_i - \underline{m}_i \cdot \delta\underline{\theta}_{i-1} = \underline{m}_i \cdot \underbrace{[\delta\underline{\theta}_i - \delta\underline{\theta}_{i-1}]}_{\underline{k}_i\,\delta q_i} = m_i\,\delta q_i$$

$m_i\,\underline{k}_i$

(*) This is the case considered in the following. The case of non congruent actuation can be reduced to external f/t case

$\boxed{TJ}$ $\underline{f}_i$ force generated by actuator i on link i

$-\underline{f}_i$ force generated by actuator i on link $i-1$

$$\delta L_i = \underline{f}_i \cdot \delta \underline{P}_i - \underline{f}_i \cdot \delta \underline{Q}_{i-1} = \underbrace{\underline{f}_i}_{f_i \underline{k}_i} \cdot \underbrace{\left[\delta \underline{P}_i - \delta \underline{Q}_{i-1} \right]}_{\underline{k}_i \delta q_i} = f_i \, \delta q_i$$

NOTE Let $\tau_i = \begin{cases} m_i & i \in RJ \\ f_i & i \in TJ \end{cases}$ then $\delta L_i = \tau_i \, \delta q_i$

τ_i: generalized joint actuation force

The total virtual work of the joint generalized f/t is

$$\delta L = \sum_{1}^{n} {}_i \, \delta L_i = \delta \underline{q}^T \underline{\tau} \qquad \underline{\tau} \triangleq \begin{bmatrix} \tau_1 \\ \vdots \\ \tau_n \end{bmatrix}$$